

ESXi Graceful SSH Shutdown & Boot Guide

Standard Operating Procedure for Automated Remote Power Management on Standalone VMware ESXi Hypervisors

1. INTRODUCTION & ARCHITECTURE OVERVIEW

Objective: Establish a reliable, automated shutdown and power-restoration workflow for standalone ESXi hosts during power events (e.g., UPS low battery) using client automation tools like Home Assistant or Network UPS Tools (NUT).

Unlike enterprise vSphere clusters managed by vCenter, a **standalone ESXi host does not require Maintenance Mode** prior to powering down. Running `esxcli system shutdown poweroff` directly triggers the native host daemon (`hostd`), which automatically coordinates guest OS shutdowns according to defined host Autostart policies.

2. ESXI HOST PREPARATION

A. Enable SSH Service

1. Log into the ESXi Web Client (Embedded Host Client).
2. Navigate to **Manage** → **Services** in the left pane.
3. Select **TSM-SSH** (SSH Service).
4. Click **Actions** → **Policy** → Set to **Start and stop with host**.
5. Start the SSH service if currently stopped.

B. Configure Global Autostart & Autostop Policy

Navigate to **Manage** → **System** → **Autostart** and click **Edit settings**:

Setting	Recommended Value	Description / Purpose
Enabled	Yes	Enables host-level autostart and autostop processing.
Start delay	120 seconds	Staggeres startup times between sequential VM groups.
Stop delay	120 seconds	Maximum duration allowed for graceful guest OS shutdown response.
Stop action	Shut down	Crucial: Sends OS shutdown signal. Do NOT use <i>Power off</i> or <i>Suspend</i> .
Wait for heartbeat	Yes	Confirms VMware Tools is active before starting the next VM group.

C. Configure Individual Virtual Machine Startup Priority

- In the **Autostart** tab, select target VMs and click **Enable**.
- Set explicit **Start Order** so infrastructure services (e.g., Active Directory, DNS, Storage) boot before application workloads.
- **Prerequisite:** Ensure **VMware Tools** (or `open-vm-tools`) is installed on every VM to process guest OS ACPI shutdown signals.

3. PASSWORDLESS SSH KEY AUTHENTICATION SETUP

To execute remote commands without exposing plain-text credentials, set up an RSA SSH key pair from your automation client (e.g., Home Assistant or UPS management box):

Step 1: Generate Key Pair on Remote Client

```
mkdir -p /config/ssh_keys ssh-keygen -t rsa -b 4096 -f /config/ssh_keys/id_rsa_homeassistant -N ""
```

Step 2: Transfer Public Key to ESXi Host

```
cat /config/ssh_keys/id_rsa_homeassistant.pub | ssh root@<ESXI_IP> 'cat >> /etc/ssh/keys-root/authorized_keys'
```

Step 3: Test Execution Command

```
ssh -i /config/ssh_keys/id_rsa_homeassistant -o StrictHostKeyChecking=no root@<ESXI_IP> "esxcli system shutdown poweroff -r 'UPS Low Battery Test'"
```

4. EXECUTION COMMAND WORKFLOW

The official, supported CLI command for shutting down standalone ESXi hosts is:

```
esxcli system shutdown poweroff -r "UPS Low Battery - Automated Shutdown"
```

- 1 Command Received:** ESXi hostd daemon receives the poweroff instruction and system reason text.
- 2 VM Guest Shutdown:** ESXi dispatches graceful guest OS shutdown signals via VMware Tools to active VMs.
- 3 Timer Enforcement:** System waits up to the configured Stop Delay for guest applications to flush writes and halt.
- 4 Timeout Fallback:** Any hung or unresponsive guest OS is forcibly stopped once the timeout expires to prevent UPS battery drain.
- 5 Host Power Off:** Storage volumes unmount safely, system caches flush, and the physical motherboard powers off.

5. PHYSICAL HARDWARE AUTOMATIC RESTORATION (BIOS/UEFI)

To ensure the host automatically powers back on and restarts VMs when grid utility power returns:

1. Enter the server physical BIOS/UEFI or Out-of-Band Management interface (Dell iDRAC, HPE iLO, Lenovo XCC).
2. Locate the **AC Power Recovery** or **Restore on AC Power Loss** configuration setting.